

TsaoSciResearcher Engineering Audit

Release 0.7.4 | composite evidence | 2026-08-07

Evidence-first scientific research control layer

Question -> evidence -> design -> execution handoff -> validation -> artifact

Truth boundary: a contract or handoff is not an execution receipt.

341

Capability contracts

322

Named catalog contracts

15

Workflows

20

Schemas

43

Test modules

7

Domain packs

Delivered integration

1. Measurement Boundary with calibration, uncertainty and applicability fields
2. Structure-Property planning with confounders, alternatives and scale bridges
3. Bilingual causality guard and evidence-to-source execution-receipt checks
4. Deterministic test/quality HTML-SVG interfaces and this four-page PDF

Repository operation

All changes were written directly to main; no branch and no pull request were created.

Scoped software validation matrix

Recorded cross-platform baseline

These platform results are a recorded baseline, not a fresh current-tree run.

macos python 3 12		PASS
ubuntu python 3 10		PASS
ubuntu python 3 13		PASS
windows python 3 12		PASS

Scoped validation gates

baseline bandit high severity		PASS
baseline bounded performance		PASS
baseline byte identical release builds		PASS
baseline complete regression		PASS
baseline critical mutation killed		24/24
baseline mypy strict		PASS
baseline repository and contract audit		PASS
baseline reverse order regression		PASS
baseline ruff format and lint		PASS
baseline seeded random order regression		PASS
current end to end ci		NOT RUN
deterministic visual report contract		PASS
focused current change regression		PASS
readme version and mirror alignment		PASS
scientific quality guard regression		PASS

Scoped status applies only to the evidence levels explicitly recorded above.

Scientific quality controls

Four deterministic contracts: measurement, multiscale reasoning, causal claims and evidence provenance.

Examples: PASS 3 | WARN 0 | BLOCK 1 | average completeness 89%

Measurement boundary **PASS**

Declared completeness: 100%



Operational definition, calibration, uncertainty and applicability are explicit.

MB-COMPLETE: Operational measurement boundary is explicit.

Kind: measurement-boundary

Structure-property planner PASS

Declared completeness: 100%



The chain records intervention, structure, mediator, response and competing...

SP-CHAIN: Structure, mediator and property are all explicit.

Kind: structure-property-plan

Causality guard **BLOCK**

Declared completeness: 57%



Association-only evidence blocks causal wording in English and Chinese.

CG-OVERCLAIM: Causal wording exceeds the declared study design.

Kind: causality-guard

Evidence traceability	PASS
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Declared completeness: 100%



Claims link to evidence locators and execution claims require receipts.

ET-COMplete: Claim, evidence, locators and execution boundary are traceable.

Kind: evidence-traceability

BLOCK is intentional when wording exceeds evidence or an execution claim lacks a receipt.

The guards remain configurable and do not hard-code material-specific trends.

Scoped evidence and limitations

Recorded validation scope

Scope: composite

Only the explicitly recorded checks may be interpreted as passing.

Current-tree end-to-end CI

Status: NOT_RUN

No exact commit attestation is asserted until external CI artifacts record it.

Explicit limitations

1. Current-tree end-to-end CI remains NOT_RUN for this acceptance hardening delta.
2. External scientific execution requires a checksum-verifiable execution receipt.

Reproduction entry points

```
python scripts/build_validation_evidence.py --check
```

```
python scripts/build_test_dashboard.py --check
```

```
python scripts/build_research_quality_dashboard.py --check
```

```
python scripts/build_engineering_report.py --check
```

Software validation is not acceptance of experiments, external calculations, legal or safety claims.